

HECTOR GENARO PACHECO

Software Engineer | Full-Stack • Backend & APIs • Cloud Infrastructure • Mobile Systems

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PROFESSIONAL SUMMARY

Versatile Software Engineer with experience delivering standalone systems and collaborative contributions in shared codebases. Background spans full-stack web platforms, containerized cloud infrastructure on AWS ECS Fargate, type-safe REST/GraphQL APIs, relational database modeling (PostgreSQL/Prisma), native Android mobile apps (Kotlin/Jetpack Compose), embedded Linux daemons, and Jira/GitLab-driven team workflows with automated testing.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C++, Kotlin

Frontend & Mobile: React, Next.js 16, Android (Kotlin), Jetpack Compose, HTML5, CSS3, Tailwind CSS

Backend & APIs: Node.js, Express, REST APIs, GraphQL, FastAPI, Zod

AI & Platforms: Palantir Foundry, Palantir AIP, OpenAI API, RAG

Cloud & DevOps: AWS (ECS Fargate, ECR, RDS, Secrets Mgr), Docker, Linux, Git, GitLab, Jira, Vercel

Data & Databases: PostgreSQL, Prisma ORM, MongoDB, Supabase

Systems & Hardware: Raspberry Pi 5, gphoto2, systemd, GPIO, 4–20 mA, PLC

Quality & Testing: Vitest, Unit & Integration Testing, Test-Driven Dev, Audit Logging

FEATURED SOFTWARE & SYSTEMS ENGINEERING PROJECTS

EstateOS — *Real Estate Investment Intelligence Platform* [CASE STUDY ↗](#) **Next.js 16 • React • TypeScript • PostgreSQL • Prisma ORM • Vitest • TailwindCSS**

- Built a full-stack real estate analysis platform featuring deterministic financial underwriting, debt capital stacks, LTV/CLTV, and equity calculations.
- Engineered multi-strategy screening (Wholesale, Fix & Flip, BRRRR, Buy & Hold) with pure TypeScript calculators and strict input validation.
- Architected identity-bound evidence snapshots (PropertyEvidenceSnapshot) and source authority classifiers, verified by an 804-test suite in Vitest.

CalTrack — *Industrial Calibration SaaS & Mobile Platform* **React • TypeScript • Node.js • Express • PostgreSQL • Docker • AWS ECS Fargate • Kotlin / Android**
[LIVE APP ↗](#) [GITHUB ↗](#)

- Architected and deployed a full-stack industrial calibration SaaS platform managing instrument tags, calibration logs, RBAC, and auditable history snapshots.
- Containerized Node/Express backend with Docker on AWS ECS Fargate, ECR, RDS PostgreSQL, Secrets Manager, KMS, and Vercel Edge.
- Engineered a native Android field companion application in Kotlin with Jetpack Compose, ViewModels, CameraX, and ML Kit barcode scanning.

Interactive Photo Booth System — *Client Hardware/Software Integration* [CASE STUDY ↗](#) **Raspberry Pi 5 • Linux • Python • photobooth-app • gphoto2 • Canon DSLR • systemd**

- Built an event photo booth system for a nonprofit client, integrating Raspberry Pi 5, active cooling, an Acer 15.6" touchscreen, and tethered Canon DSLR.
- Automated camera capture via gphoto2 over USB PTP and configured systemd user services with kiosk startup scripts for reliable event operation.

Additional Repositories & Deployments: **Procurement Tracking** — Full-Stack GraphQL App (React/Apollo/Prisma/PostgreSQL) [LIVE APP ↗](#) [GITHUB ↗](#) • **FieldTrack AI** — Edge Telemetry (Raspberry Pi/Python/React) [GITHUB ↗](#) • **OC Water Features** — Production Web (Next.js/Supabase/Vercel) [LIVE SITE ↗](#)

TECHNICAL & INDUSTRIAL EXPERIENCE

Software Engineering Contributor — *Nonprofit serving underserved communities* **Collaborative Contribution**

- Contributed to an existing nonprofit software platform using Jira-based development workflows, Git/GitLab version control, API investigation, and collaborative debugging within a shared codebase.

Instrumentation & Controls Technician — *Process Control & Industrial Systems* **Field Engineering Experience**

- Calibrated, commissioned, and troubleshot 4–20 mA process instrumentation, PLC I/O wiring, relays, and field sensors across 480VAC, 120VAC, and 24VDC power systems with systematic fault isolation and auditable logs.

EDUCATION & PROFESSIONAL TRAINING

Western Governors University (WGU) — B.S. Software Engineering (In Progress, 2026–Present)

Palantir Technologies — Foundry & AIP Learning Credentials (May 2026)

Mirion Technologies — SU-890 iCAM Operations & Maintenance (March 2026, 16.00 Continued Education Credits)